Phase 2 - Week 6 - Lesson Day 4

Model Comparison + Week 6 Mini Challenge

Today's Goal

Today you put the whole week together. You will train three models on the same data and compare them fairly, using the metrics from Day 3.

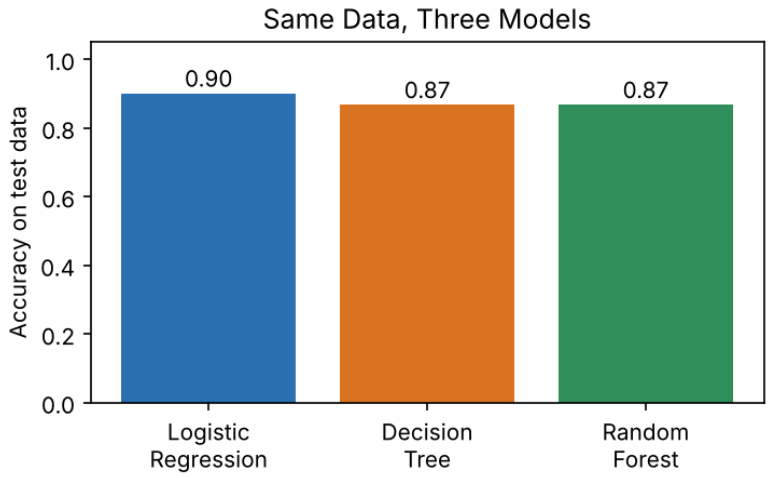
- Review the three classification models you know
- Compare models fairly
- Pick a model and explain why, in plain English

1. Quick Review

Model	How it works	Good to know
Logistic Regression	Draws one smooth boundary using a formula	Simple and fast. Great when the pattern is fairly straight.
Decision Tree	Asks a chain of yes/no questions	Easy to read and explain. Overfits if you do not limit depth.
Random Forest	Many trees vote together	Usually steady and strong. Harder to explain than one tree.

2. Comparing Models Fairly

The fair-test rules: (1) use the same training data for every model, (2) use the same test data for every model, (3) use the same metrics, and (4) judge on the test data, never the training data.



A surprise worth remembering: in our student data, Logistic Regression scored as well as, or better than, the tree and the forest. That is because we built the data so that passing depends on a fairly straight-line mix of hours and attendance, and that is exactly what Logistic Regression is good at. The fancier model does not always win. Always test, then choose.

Also be careful with small test sets: we only test on 60 students, so one student changes the score by about 0.017. A gap of 0.03 between models is only about 2 students, which could easily be luck. Treat close scores as a tie, and then pick the simplest model.

3. A Simple Comparison Loop

```
models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "Decision Tree": DecisionTreeClassifier(max_depth=3, random_state=42),
    "Random Forest": RandomForestClassifier(n_estimators=100, random_state=42),
}

for name, model in models.items():
    model.fit(X_train, y_train)               # same training data
    predictions = model.predict(X_test)       # same test data
    print(name, accuracy_score(y_test, predictions))
```

Putting models in a dictionary and looping over them means you write the code once and it works for any number of models.

Week 6 Mini Challenge

Open **day-4-model-comparison-mini-challenge.ipynb**. It has a fully solved mini challenge you can also reuse as a template on your own dataset.

Mini Challenge Steps

1. Create the student dataset and split it into training and test sets.
2. Train three models: Logistic Regression, Decision Tree (max_depth=3), and Random Forest.
3. For each model, calculate accuracy, precision, recall, F1, and ROC-AUC on the test data.
4. Put all the results into one table (a DataFrame).
5. Draw a bar chart comparing the F1 scores.
6. Write a short conclusion: which model would you pick, and why? Remember that the simplest model that does the job is often the best choice.
7. Use your chosen model to predict for a new student.

Once you and Halimo finish, Week 6 is complete! Next up is Week 7: Model Optimization (cross-validation, tuning, and regularization).